The transition matrix element of arXiv:2607.02812 in helicity space

Rashba–Hubbard superconductor: quasiparticle scattering and pair creation, helicity vertices for ultrasound and electromagnetic absorption, and an exact numerical check

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The paper: R. Afzali, B. Takrim, A. H. Asghari, *Theoretical investigation of power absorbed by superconductors with Rashba spin-orbit coupling in the presence external time-dependent perturbation*, arXiv:2607.02812 (2026). Its model, helicity basis and Bogoliubov transformation are those of J. Hutchinson, J. E. Hirsch, F. Marsiglio, Phys. Rev. B **97**, 184513 (2018), arXiv:1801.08988 ("HHM", "HHM Eq. n"). The paper's formulas carry no equation numbers in the version I read, so they are quoted by name; equations of this note are labelled (1), (2), …

Result. In the helicity basis the paper's matrix element keeps its form ($\alpha \to s$; unprimed $= (\mathbf{k}, s)$, primed $= (\mathbf{k}', s')$),

$$M(\mathbf{k}s|\mathbf{k}'s') = B(\mathbf{k}s|\mathbf{k}'s')\left[u_{\mathbf{k}s}u_{\mathbf{k}'s'} - \eta\, v_{\mathbf{k}s}v_{\mathbf{k}'s'}\right] \tag{M}$$

with three points that the constant-B treatment misses. (i) $B(\mathbf{k}s|\mathbf{k}'s') = \langle \mathbf{k}s|\widehat{O}|\mathbf{k}'s'\rangle$ is the *helicity-space vertex* of the perturbation $\widehat{O}$; it is not a constant (Sec. 3). (ii) $u_{\mathbf{k}s} = \sqrt{\tfrac{1}{2}(1 + \xi_{\mathbf{k}s}/E_{\mathbf{k}s})}$ and $v_{\mathbf{k}s} = \mathrm{sgn}(\bar{\Delta}_{\mathbf{k}s})\sqrt{\tfrac{1}{2}(1 - \xi_{\mathbf{k}s}/E_{\mathbf{k}s})}$ are real, with $\xi_{\mathbf{k}s} = \varepsilon_{\mathbf{k}s} - \mu$, $E_{\mathbf{k}s} = \sqrt{\xi^2_{\mathbf{k}s} + \bar{\Delta}^2_{\mathbf{k}s}}$ and the *signed* gap $\bar{\Delta}_{\mathbf{k}s} = \Delta^0 - \Delta^s\varepsilon_{\mathbf{k}s}/4t$ (HHM Eq. 37). (iii) The time-reversal phases are those of the helicity states, $\rho_{\mathbf{k}s} = s\, e^{-i\theta_{\mathbf{k}}}$; they cancel exactly against the phase of the gap $\Delta_{\mathbf{k}s} = \rho_{\mathbf{k}s}\bar{\Delta}_{\mathbf{k}s}$. $\eta = +1$ for a time-reversal-even operator (ultrasound, the paper's case I), $\eta = -1$ for a time-reversal-odd one (current, case II).

$$|M|^2 = |B(\mathbf{k}s|\mathbf{k}'s')|^2\,\frac{1}{2}\left[1 + \frac{\xi\xi'}{EE'} - \eta\,\frac{\bar{\Delta}\bar{\Delta}'}{EE'}\right] \tag{C}$$

Pair creation (the other two terms of the paper's H'), with $X = \langle \mathbf{k}s|\widehat{O}|T(\mathbf{k}'s')\rangle$, T = time reversal:

$$N(\mathbf{k}s, \mathbf{k}'s') = X\left[u_{\mathbf{k}s}v_{\mathbf{k}'s'} + \eta\, u_{\mathbf{k}'s'}v_{\mathbf{k}s}\right], \qquad |N|^2 = |X|^2\,\frac{1}{2}\left[1 - \frac{\xi\xi'}{EE'} + \eta\,\frac{\bar{\Delta}\bar{\Delta}'}{EE'}\right] \tag{N}$$

Vertices (Sec. 3). Ultrasound, spin-independent coupling of strength B_q, $\eta = +1$: $|B|^2 = |B_q|^2\,\frac{1}{2}\left(1 + ss'\cos\Delta\theta\right)$, $\Delta\theta = \theta_{\mathbf{k}} - \theta_{\mathbf{k}'}$. Electromagnetic absorption at $q \to 0$, $\eta = -1$: $B = \mathbf{A}\cdot\mathbf{V}_{ss'}(\mathbf{k})$ with $\mathbf{V}_{ss} = \nabla_{\mathbf{k}}\varepsilon_{\mathbf{k}s}$ and an interband part $V^x_{s,-s} = 2iV_{SO}\, s\cos k_x \sin k_y/R_{\mathbf{k}}$, $V^y_{s,-s} = -2iV_{SO}\, s\cos k_y \sin k_x/R_{\mathbf{k}}$, $R_{\mathbf{k}} = \sqrt{\sin^2 k_x + \sin^2 k_y}$.

1. Setting and conventions

The non-interacting part is $H_0(\mathbf{k}) = \varepsilon_{\mathbf{k}} - \mu + H_{SO}(\mathbf{k})$ with $\varepsilon_{\mathbf{k}} = -2t(\cos k_x + \cos k_y)$ and the Rashba term of the paper (HHM Eq. 14),

$$H_{SO}(\mathbf{k}) = -2V_{SO}\left(\sin k_y\,\sigma_x - \sin k_x\,\sigma_y\right) = -2V_{SO}R_{\mathbf{k}}\,\hat{\mathbf{g}}_{\mathbf{k}}\cdot\sigma, \qquad \hat{\mathbf{g}}_{\mathbf{k}} = (\cos\theta_{\mathbf{k}}, \sin\theta_{\mathbf{k}}) \tag{1}$$

Its eigenstates are the helicity states (HHM Eqs. 15–17), $e^{i\theta_{\mathbf{k}}} = (\sin k_y - i\sin k_x)/R_{\mathbf{k}}$:

$$|\mathbf{k}s\rangle = \frac{1}{\sqrt{2}}\left(1,\ s\,e^{i\theta_{\mathbf{k}}}\right)^T, \qquad \varepsilon_{\mathbf{k}s} = \varepsilon_{\mathbf{k}} - 2sV_{SO}R_{\mathbf{k}}, \qquad s = \pm 1 \tag{2}$$

Time reversal $T = i\sigma_y K$ keeps the helicity and maps $\mathbf{k} \to -\mathbf{k}$; since $e^{i\theta_{-\mathbf{k}}} = -e^{i\theta_{\mathbf{k}}}$,

$$T|\mathbf{k}s\rangle = \rho_{\mathbf{k}s}|-\mathbf{k}, s\rangle, \qquad \rho_{\mathbf{k}s} = s\,e^{-i\theta_{\mathbf{k}}}, \qquad \rho_{-\mathbf{k}s} = -\rho_{\mathbf{k}s}, \qquad T^2 = -1 \tag{3}$$

(HHM write the time-reversed partner of $c_{\mathbf{k}s}$ as $-s\,e^{i\theta}c_{-\mathbf{k}s}$: the same up to the overall sign convention of T.)
The gap. Pairing is intraband, between $|\mathbf{k}s\rangle$ and its time-reverse, and HHM (text after Eq. 22) find

$$\Delta_{\mathbf{k}s} = s\,e^{-i\theta_{\mathbf{k}}}\bar{\Delta}_{\mathbf{k}s} = \rho_{\mathbf{k}s}\bar{\Delta}_{\mathbf{k}s}, \qquad \bar{\Delta}_{\mathbf{k}s} = \Delta^0 + \Delta^s s_{\mathbf{k}} + \Delta^{x-y}sR_{\mathbf{k}} = \Delta^0 - \frac{\Delta^s}{4t}\varepsilon_{\mathbf{k}s} \quad \text{(real)} \tag{4}$$

In spin space this is a singlet $\Delta^0 + \Delta^s s_{\mathbf{k}}$ plus a triplet with $\mathbf{d}_{\mathbf{k}} = \Delta^{x-y}(\sin k_y, -\sin k_x, 0)$ along the Rashba vector, $\Delta^{x-y} = V_{SO}\Delta^s/2t$ (HHM Eq. 35). I checked numerically that this spin-space gap matrix projects onto *exactly* the form (4), diagonal in s (Sec. 5). **Bogoliubov transformation** (HHM Eq. 23), for the helicity operators $a_{\mathbf{k}s}$ ($a^\dagger_{\mathbf{k}s}$ creates $|\mathbf{k}s\rangle$):

$$a_{\mathbf{k}s} = u_{\mathbf{k}s}\,\gamma_{\mathbf{k}s} - \rho_{\mathbf{k}s}\,v_{\mathbf{k}s}\,\gamma^\dagger_{-\mathbf{k}s}, \qquad u_{\mathbf{k}s}v_{\mathbf{k}s} = \frac{\bar{\Delta}_{\mathbf{k}s}}{2E_{\mathbf{k}s}}, \qquad u^2_{\mathbf{k}s} - v^2_{\mathbf{k}s} = \frac{\xi_{\mathbf{k}s}}{E_{\mathbf{k}s}} \tag{5}$$

u, v are real and even in $\mathbf{k}$; the oddness of the gap sits in $\rho_{\mathbf{k}s}$. In HHM's convention the pairing term is $-\frac{1}{2}\Delta c^\dagger c^\dagger$ and v has the opposite overall sign; only products vv' enter below.

2. Derivation

The perturbation is a one-body operator, the paper's $H' = \sum B(\mathbf{k}\alpha|\mathbf{k}'\alpha')c^\dagger_{\mathbf{k}\alpha}c_{\mathbf{k}'\alpha'}$; in the helicity basis

$$H' = \sum_{\mathbf{k}\mathbf{k}'}\sum_{ss'} B(\mathbf{k}s|\mathbf{k}'s')\,a^\dagger_{\mathbf{k}s}a_{\mathbf{k}'s'}, \qquad B(\mathbf{k}s|\mathbf{k}'s') = \langle\mathbf{k}s|\hat{O}|\mathbf{k}'s'\rangle \tag{6}$$

Step 1 — insert (5).

$$a^\dagger_{\mathbf{k}s}a_{\mathbf{k}'s'} = u\,u'\,\gamma^\dagger_{\mathbf{k}s}\gamma_{\mathbf{k}'s'} + \rho^*_{\mathbf{k}s}\rho_{\mathbf{k}'s'}\,v\,v'\,\gamma_{-\mathbf{k}s}\gamma^\dagger_{-\mathbf{k}'s'} - u\,\rho_{\mathbf{k}'s'}v'\,\gamma^\dagger_{\mathbf{k}s}\gamma^\dagger_{-\mathbf{k}'s'} - \rho^*_{\mathbf{k}s}v\,u'\,\gamma_{-\mathbf{k}s}\gamma_{\mathbf{k}'s'} \tag{7}$$

Step 2 — the scattering amplitude.

The second term of (7) is $-\rho^*_{\mathbf{k}s}\rho_{\mathbf{k}'s'}vv'\,\gamma^\dagger_{-\mathbf{k}'s'}\gamma_{-\mathbf{k}s}$ plus a constant. Relabelling $(\mathbf{k}, s, \mathbf{k}', s') \to (-\mathbf{k}', s', -\mathbf{k}, s)$, using $\rho_{-\mathbf{k}s} = -\rho_{\mathbf{k}s}$ and the evenness of v, the coefficient of $\gamma^\dagger_{\mathbf{k}s}\gamma_{\mathbf{k}'s'}$ is

$$M(\mathbf{k}s|\mathbf{k}'s') = B(\mathbf{k}s|\mathbf{k}'s')\,u_{\mathbf{k}s}u_{\mathbf{k}'s'} - \rho_{\mathbf{k}s}\rho^*_{\mathbf{k}'s'}\,v_{\mathbf{k}s}v_{\mathbf{k}'s'}\,B(-\mathbf{k}'s'|-\mathbf{k}s) \tag{8}$$

This is the paper's general expression $M = B\,u_k u_{k'} - v_k v_{k'}\sum\rho\rho\,B(-k'\alpha'|-k\alpha)$ with the spin phases replaced by the helicity phases.

Step 3 — time-reversal lemma.

Let $\hat{O}$ be Hermitian with $T\hat{O}T^{-1} = \eta\hat{O}$. For an antiunitary T, $\langle T\phi|\hat{O}|T\psi\rangle = \eta\langle\psi|\hat{O}|\phi\rangle$. With $|-\mathbf{k}s\rangle = \rho^*_{\mathbf{k}s}\,T|\mathbf{k}s\rangle$ from (3) this gives

$$B(-\mathbf{k}'s'|-\mathbf{k}s) = \eta\,\rho_{\mathbf{k}'s'}\rho^*_{\mathbf{k}s}\,B(\mathbf{k}s|\mathbf{k}'s') \tag{9}$$

This is the paper's relation $\sum \rho\rho\, B(-k'\alpha'|-k\alpha) = \eta B(k\alpha|k'\alpha')$, again with helicity phases: $\eta = +1$ for a scalar (density-like) operator, $\eta = -1$ for a current-like one.

Step 4 — the phases cancel.

Insert (9) into (8): the phase factor is $\rho_{\mathbf{k}s}\rho^{*}_{\mathbf{k}'s'} \cdot \rho_{\mathbf{k}'s'}\rho^{*}_{\mathbf{k}s} = 1$, so the second term of (8) is $-\eta\, B\, v_{\mathbf{k}s} v_{\mathbf{k}'s'}$ and (M) follows. This works *because* the gap carries exactly the phase $\rho_{\mathbf{k}s}$ (4), which makes v real. Had the phase of the gap been dropped, or $|\Delta_{\mathbf{k}s}|$ used, a factor like $e^{i(\theta_{\mathbf{k}} - \theta_{\mathbf{k}'})}$ would survive in the coherence term.

Step 5 — square it.

With $u^2 = \frac{1}{2}(1 + \xi/E)$, $v^2 = \frac{1}{2}(1 - \xi/E)$ and $uv = \bar{\Delta}/2E$ from (5):

$$(uu' - \eta vv')^2 = u^2u'^2 + v^2v'^2 - 2\eta\, uu'vv' = \frac{1}{2}\left[1 + \frac{\xi\xi'}{EE'} - \eta\frac{\bar{\Delta}\bar{\Delta}'}{EE'}\right] \tag{10}$$

which is (C). Because $\bar{\Delta}$ changes sign at the node $\varepsilon_0 = 4t\Delta^0/\Delta^s$, the product $\bar{\Delta}\bar{\Delta}'$ must keep its sign.

Step 6 — pair creation.

The two cross terms of (7) create or annihilate pairs of quasiparticles. Antisymmetrising the creation term and using $\langle\psi|\widehat{O}|T\phi\rangle = -\eta\langle\phi|\widehat{O}|T\psi\rangle$ (from the lemma and $T^2 = -1$) gives the amplitude $N_{ab} = X_{ab}(u_a v_b + \eta\, u_b v_a)$ for $\gamma^\dagger_a\gamma^\dagger_b$, $X_{ab} = \langle a|\widehat{O}|Tb\rangle$, and $|N|^2$ as in (N).

3. The helicity vertex B(ks|k's')

3.1 Ultrasound: spin-independent coupling, η = +1

A deformation-potential coupling $\widehat{O} = B_q\rho_{\mathbf{q}} + \mathrm{h.c.}$ is the identity in spin space, so $B(\mathbf{k}s|\mathbf{k}'s') = B_q\langle\mathbf{k}s|\mathbf{k}'s'\rangle$ with $\mathbf{k} = \mathbf{k}' + \mathbf{q}$, and from (2)

$$\langle\mathbf{k}s|\mathbf{k}'s'\rangle = \frac{1}{2}\left(1 + ss'\, e^{i(\theta_{\mathbf{k}'} - \theta_{\mathbf{k}})}\right), \qquad |\langle\mathbf{k}s|\mathbf{k}'s'\rangle|^2 = \frac{1}{2}\left(1 + ss'\cos\Delta\theta\right), \quad \cos\Delta\theta = \hat{\mathbf{g}}_{\mathbf{k}}\cdot\hat{\mathbf{g}}_{\mathbf{k}'} \tag{11}$$

Together with (C), the complete ultrasonic matrix element for scattering is

$$|M(\mathbf{k}s|\mathbf{k}'s')|^2 = \frac{|B_q|^2}{4}\left(1 + ss'\cos\Delta\theta\right)\left[1 + \frac{\xi\xi'}{EE'} - \frac{\bar{\Delta}\bar{\Delta}'}{EE'}\right] \tag{12}$$

For pair creation ($\mathbf{k} + \mathbf{k}' = \mathbf{q}$) the overlap is with $|-\mathbf{k}'s'\rangle$, and since $\theta_{-\mathbf{k}} = \theta_{\mathbf{k}} + \pi$ the roles of intra- and inter-helicity are exchanged: $|X|^2 = |B_q|^2\frac{1}{2}(1 - ss'\cos\Delta\theta)$.

Selection rule. For $q \ll \pi/a$ the Rashba vectors at $\mathbf{k}$ and $\mathbf{k}' = \mathbf{k} - \mathbf{q}$ are nearly parallel: the weight is ≈ 1 for $s' = s$ and ≈ 0 for $s' = -s$. The Brillouin-zone average of the two weights (independent of V_{SO}, since only $\theta_{\mathbf{k}}$ enters), for $\mathbf{q}$ along k_x, is

q (units of $1/a$)	0.05	0.10	0.30	1.00	2.00	3.14
intra-helicity, $s' = s$	0.9990	0.9966	0.9777	0.8514	0.6396	0.5000
inter-helicity, $s' = -s$	0.0010	0.0034	0.0223	0.1486	0.3604	0.5000

Only for $q \sim \pi/a$ does the weight reach the value $\frac{1}{2}$ in all four (s, s') channels, which is what a constant B assumes (Sec. 4).

3.2 Electromagnetic absorption: current vertex, $\eta = -1$

In the dipole limit ($q \to 0$) the coupling is $\mathbf{A}\cdot\widehat{\mathbf{V}}$ with the velocity operator $\mathbf{V} = \partial_{\mathbf{k}} H_0$ (all constants absorbed in $\mathbf{A}$). It is odd under T and diagonal in $\mathbf{k}$:

$$V_x = 2t\sin k_x + 2V_{SO}\cos k_x\,\sigma_y, \qquad V_y = 2t\sin k_y - 2V_{SO}\cos k_y\,\sigma_x \tag{13}$$

With (2), $\langle \mathbf{k}s|\sigma_x|\mathbf{k}s'\rangle = s\cos\theta_{\mathbf{k}}$ and $\langle \mathbf{k}s|\sigma_y|\mathbf{k}s'\rangle = s\sin\theta_{\mathbf{k}}$ for $s' = s$, and $-i\,s\sin\theta_{\mathbf{k}}$, $i\,s\cos\theta_{\mathbf{k}}$ for $s' = -s$; with $\cos\theta_{\mathbf{k}} = \sin k_y/R_{\mathbf{k}}$, $\sin\theta_{\mathbf{k}} = -\sin k_x/R_{\mathbf{k}}$ this gives

$$V^{x}_{ss} = 2t\sin k_x - 2sV_{SO}\frac{\sin k_x\cos k_x}{R_{\mathbf{k}}} = \partial_{k_x}\varepsilon_{\mathbf{k}s}, \qquad V^{x}_{s,-s} = 2iV_{SO}\,s\,\frac{\cos k_x\sin k_y}{R_{\mathbf{k}}} \tag{14}$$

$$V^{y}_{ss} = 2t\sin k_y - 2sV_{SO}\frac{\sin k_y\cos k_y}{R_{\mathbf{k}}} = \partial_{k_y}\varepsilon_{\mathbf{k}s}, \qquad V^{y}_{s,-s} = -2iV_{SO}\,s\,\frac{\cos k_y\sin k_x}{R_{\mathbf{k}}} \tag{15}$$

The interband element equals $(\varepsilon_{\mathbf{k},-s} - \varepsilon_{\mathbf{k}s})\langle \mathbf{k}s|\,\partial_{\mathbf{k}}|\mathbf{k},-s\rangle$ with $\varepsilon_{\mathbf{k},-s} - \varepsilon_{\mathbf{k}s} = 4sV_{SO}R_{\mathbf{k}}$. From (M) and (N) with $\eta = -1$, $\mathbf{k}' = \mathbf{k}$:

$$M(\mathbf{k}s|\mathbf{k}s') = \mathbf{A}\cdot\mathbf{V}_{ss'}\,(\,u_{\mathbf{k}s}u_{\mathbf{k}s'} + v_{\mathbf{k}s}v_{\mathbf{k}s'}\,), \qquad N(\mathbf{k}s,-\mathbf{k}s') = \mathbf{A}\cdot\mathbf{V}_{ss'}\,(\,u_{\mathbf{k}s}v_{\mathbf{k}s'} - u_{\mathbf{k}s'}v_{\mathbf{k}s}\,) \tag{16}$$

- **Intraband** ($s' = s$): $M = \mathbf{A}\cdot\nabla\varepsilon_{\mathbf{k}s}$ (since $u^2 + v^2 = 1$), but there is no energy transfer, so no absorption at $\omega \neq 0$; and $N(\mathbf{k}s,-\mathbf{k}s) \equiv 0$ — no pair breaking within one band at $q = 0$ (BdG check: exactly zero).
- **Interband** ($s' = -s$): scattering when $E_{\mathbf{k},-s} - E_{\mathbf{k}s} = \hbar\omega$, and pair creation above $E_{\mathbf{k}+} + E_{\mathbf{k}-}$. Both are generically non-zero and are driven by V_{SO} through the interband vertex.

So at $q = 0$ the clean Rashba superconductor absorbs only through interband channels, which a constant-B treatment cannot represent. The paper's double integral over independent $\mathbf{k}, \mathbf{k}'$ corresponds instead to momentum-non-conserving scattering, where the weights of Sec. 3.1 apply.

4. How this differs from the formulas in the paper

1. **Constant B.** The paper averages B over angles and spin indices and takes it constant. In helicity space this replaces $|\langle \mathbf{k}s|\mathbf{k}'s'\rangle|^2$ by its uncorrelated-angle average $\frac{1}{2}$ in all four channels. That is reasonable only if $\mathbf{k}$ and $\mathbf{k}'$ are uncorrelated; for small q it misplaces all the weight of the $s' = -s$ channels (table above, Fig. 1b).

2. **Sign of the η term.** With $M = B(uu' - \eta vv')$, (10) gives $-\eta\bar{\Delta}\bar{\Delta}'/EE'$. The paper's intermediate result is the $\eta = +1$ case, $\frac{1}{2}(1 + \xi\xi'/EE' - \Delta\Delta'/EE')$, but its final expressions for W and W/W_N, as I read them in the extracted text, carry $+\eta\Delta\Delta'/EE'$; with $\eta = +1$ for ultrasound that exchanges the coherence factors of the two cases. Please check this against the typeset PDF.

3. **Signed gap.** $\bar{\Delta}_{\mathbf{k}s}$ changes sign at $\varepsilon_0 = 4t\Delta^0/\Delta^s = 0.879\,t$ in the example below. In $\bar{\Delta}\bar{\Delta}'$ the sign must be kept; $|\Delta_{\mathbf{k}s}|$ is wrong for transitions across the node.

4. **Sign of ξ in energy variables.** The paper writes $\varepsilon_{\mathbf{k}s} - \mu = +\sqrt{E^2 - \Delta^2}$ in the coherence factor. At a given E there are two roots $\varepsilon_{\pm}(E)$ (previous note) and ξ can have either sign, so

$$C_{\eta}(E,\sigma;E',\sigma') = \frac{1}{2}\left[1 + \frac{\xi_{\sigma}(E)\,\xi_{\sigma'}(E')}{E\,E'} - \eta\,\frac{\Delta_{\sigma}(E)\,\Delta_{\sigma'}(E')}{E\,E'}\right], \qquad \xi_{\sigma}(E) = \varepsilon_{\sigma}(E) - \mu \tag{17}$$

with $\varepsilon_{\sigma}(E)$, $\Delta_{\sigma}(E)$ and the band test of the $\Delta(E)$ note. **Example** ($V_{SO} = 0.5\,t$, $\Delta^0 = 0.29255\,t$, $\Delta^s = 1.33119\,t$, $\mu = 3.72649\,t$, $W = 4.2426\,t$), pairs of states with $E \approx 1.10\,t$ and $E' \approx 2.00\,t$. At E the roots are $\varepsilon_+ = 4.044\,t$ ($\xi > 0$) and $\varepsilon_- = 2.841\,t$ ($\xi < 0$); at E' they are $\varepsilon_- = 1.747\,t$ and $\varepsilon_+ = 5.138\,t > W$

, which the band test discards. I collected all pairs of grid states (300×300×2 grid, windows $\pm 0.004\,t$) in these energy windows and computed their coherence factors directly:

roots (σ,σ')	sgn $(\xi\xi')$	helicity of the states on σ	$C_{\eta=+1}$ from (17) [range on the grid]	$C_{\eta=-1}$ from (17) [range on the grid]	"+√" rule (η=+1, η=−1)
(+, −)	−1	$s=-1$ only (156 states)	0.2881 [0.2857, 0.2907]	0.4264 [0.4236, 0.4291]	**0.5736, 0.7119**
(−, −)	+1	both: $s=+1$: 72, $s=-1$: 168	0.8552 [0.8533, 0.8573]	0.9410 [0.9396, 0.9423]	**0.8552, 0.9410**
(+, +) (−, +)	–	$\varepsilon_+(E')=5.138\,t>W$: discarded by the band test; no grid state with $E\approx E'$ lies on that root (0 found)			–

The formula (17) reproduces the grid values; the "+√" prescription is right only when $\xi\xi'>0$ and is off by ≈ 0.29 in the $(+,-)$ channel, where $\xi\xi'<0$. Helicity refines the band test: the $s=+1$ band ends at $4t$ and the $s=-1$ band at W, so a root with $4t<\varepsilon\le W$ can only belong to $s=-1$. Here the $+$ root at $E=1.10\,t$ has $\varepsilon_+=4.044\,t>4t$ and indeed all its states have $s=-1$; the $+$ root hosts $s=+1$ states only for $E<E(4t)=1.074\,t$.

5. Exact numerical check

The check does not use any helicity argument. I build the 4×4 Bogoliubov–de Gennes matrix in *spin* space, in the Nambu basis $(c_{\mathbf{k}\uparrow}, c_{\mathbf{k}\downarrow}, c^\dagger_{-\mathbf{k}\uparrow}, c^\dagger_{-\mathbf{k}\downarrow})$, with the physical gap matrix (singlet + triplet along the Rashba vector):

$$H_{BdG}(\mathbf{k})=\begin{pmatrix} h(\mathbf{k}) & D(\mathbf{k}) \\ D^\dagger(\mathbf{k}) & -h^T(-\mathbf{k}) \end{pmatrix}, \qquad D(\mathbf{k})=\left[\Delta^0+\Delta^s s_{\mathbf{k}}+\Delta^{x-y}R_{\mathbf{k}}\,\hat{\mathbf{g}}_{\mathbf{k}}\cdot\sigma\right] i\sigma_y \tag{18}$$

I diagonalise it numerically and read off the coefficients of $\gamma^\dagger\gamma$ and $\gamma^\dagger\gamma^\dagger$ in $\sum Bc^\dagger c$ from the Nambu form $\mathrm{diag}(B_{\mathbf{k},\mathbf{k}'}, -B^T_{-\mathbf{k}',-\mathbf{k}})$. These are compared with the helicity-space formulas for random $(\mathbf{k},\mathbf{k}',s,s')$, with the scalar vertex and with the current vertex (two field directions).

check	max deviation
$T\lvert\mathbf{k}s\rangle=\rho_{\mathbf{k}s}\lvert-\mathbf{k}s\rangle$, eq. (3)	3×10^{-16}
spin-space gap projected on helicity states $=\delta_{ss'}\rho_{-\mathbf{k}s}\bar{\Delta}_{\mathbf{k}s}$ (no interband gap), eq. (4)	7×10^{-16}
eigenvalues of the spin-space BdG matrix $=\pm E_{\mathbf{k}\pm}$	6×10^{-15}
overlap (11) and current vertices (14)–(15) against spinor sandwiches	7×10^{-16}, 1×10^{-15}
intraband $M(\mathbf{k}s\lvert\mathbf{k}s)=\mathbf{A}\cdot\nabla\varepsilon_{\mathbf{k}s}$ against BdG	7×10^{-15}
scalar (ultrasound, η=+1): $\lvert M\rvert_{\mathrm{BdG}}-\lvert M\rvert_{\mathrm{formula}}$; $\lvert N\rvert_{\mathrm{BdG}}-\lvert N\rvert_{\mathrm{formula}}$	6×10^{-14}; 1×10^{-13}
current A at 0.0 rad (EM, η=-1): $\lvert M\rvert_{\mathrm{BdG}}-\lvert M\rvert_{\mathrm{formula}}$; $\lvert N\rvert_{\mathrm{BdG}}-\lvert N\rvert_{\mathrm{formula}}$	1×10^{-13}; 1×10^{-13}
current A at 0.9 rad (EM, η=-1): $\lvert M\rvert_{\mathrm{BdG}}-\lvert M\rvert_{\mathrm{formula}}$; $\lvert N\rvert_{\mathrm{BdG}}-\lvert N\rvert_{\mathrm{formula}}$	9×10^{-13}; 1×10^{-12}

6000 random $(\mathbf{k},\mathbf{k}',s,s')$ combinations per quantity; the largest $|M|^2$ is 8.56 in units $|B_q|=|\mathbf{A}|=1$. The BdG parameters are those of my solution of the gap equations ($U=90t$, $\Delta t=4.5t$, $n=1.875$, $k_BT=0.01t$, $V_{SO}=0.5\,t$); the identity does not depend on them. Repeating the whole check at two other parameter sets, $(V_{SO},\Delta^0,\Delta^s,\mu)/t=$ (0.90, −0.35, 2.10, −1.30) and (0.20, 0.80, −0.90, 2.20) (strong spin-orbit coupling with a negative Δ^0 and μ below the band centre; weak coupling with a negative Δ^s), gives maximum deviations in $|M|$, $|N|$ of 2×10^{-13} and 4×10^{-13}, and of at most 1×10^{-14} in all the other checks.

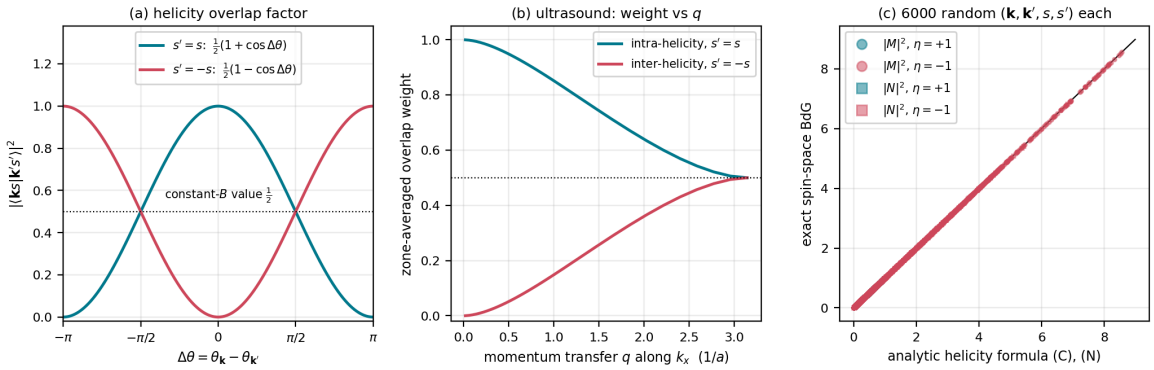


Fig. 1. (a) Helicity overlap factor (11): intra-helicity $\frac{1}{2}(1+\cos\Delta\theta)$, inter-helicity $\frac{1}{2}(1-\cos\Delta\theta)$; a constant B corresponds to the angular average $\frac{1}{2}$. (b) Brillouin-zone average of the two weights against the momentum transfer q. (c) $|M|^2$ and $|N|^2$ from the exact spin-space BdG diagonalisation against the analytic helicity formula (C), (N), for both η.

6. Minimal code and files

```
import numpy as np

def spinor(k, s):                           # |ks> = (1, s e^{i theta})/sqrt2, HHM Eqs. 16-17
    R = np.hypot(np.sin(k[0]), np.sin(k[1]))
    return np.array([1.0, s*(np.sin(k[1]) - 1j*np.sin(k[0]))/R]) / np.sqrt(2)

def uv(k, s, D0, Ds, mu, VSO, t=1.0):       # u, v (v has the sign of Dbar), E, xi, Dbar
    eps = -2*t*(np.cos(k[0]) + np.cos(k[1])) - 2*s*VSO*np.hypot(np.sin(k[0]), np.sin(k[1]))
    xi, Db = eps - mu, D0 - Ds*eps/(4*t)    # Dbar_ks = D0 - Ds eps_ks / 4t  (signed)
    E = np.hypot(xi, Db)
    return np.sqrt((1 + xi/E)/2), np.sign(Db)*np.sqrt((1 - xi/E)/2), E, xi, Db

def M_helicity(k, s, kp, sp, vertex, eta, D0, Ds, mu, VSO, t=1.0):
    """M(ks|k's') = <ks|O|k's'> (u u' - eta v v').
    vertex(k, kp): 2x2 spin matrix of O;  eta = +1 (T-even) or -1 (T-odd)."""
    u,  v  = uv(k,  s,  D0, Ds, mu, VSO, t)[:2]
    up, vp = uv(kp, sp, D0, Ds, mu, VSO, t)[:2]
    B = np.vdot(spinor(k, s), vertex(k, kp) @ spinor(kp, sp))
    return B*(u*up - eta*v*vp)

def vertex_scalar(k, kp):                   # ultrasound: spin independent, eta = +1
    return np.eye(2)

def vertex_current(k, kp, A, VSO, t=1.0):   # EM, q -> 0: A . dH/dk at (k+k')/2, eta = -1
    km = ((k[0] + kp[0])/2, (k[1] + kp[1])/2)
    sx, sy = np.array([[0, 1], [1, 0]]), np.array([[0, -1j], [1j, 0]])
    Vx = 2*t*np.sin(km[0])*np.eye(2) + 2*VSO*np.cos(km[0])*sy
    Vy = 2*t*np.sin(km[1])*np.eye(2) - 2*VSO*np.cos(km[1])*sx
    return A[0]*Vx + A[1]*Vy
```

The listing was executed against the module when this note was built and agrees with `helicity_matrix_element.py` on 300 random points for both vertices (max difference: exactly 0).

Scope. One-body perturbation; single-quasiparticle and pair-creation amplitudes at the level of the paper (no vertex corrections, screening or impurity matrix elements). M is defined up to the phase convention of $|\mathbf{k}s\rangle$; only $|M|^2$ is physical. Energy conservation and the $\mathbf{k},\mathbf{k}'$ sums that turn $|M|^2$ into the absorbed power W are not part of this note.

Files (`/home/user/deltaE/`): `Helicity_matrix_element.pdf/.html`, `helicity_matrix_element.py`, `helicity_listing.py`, `fig5_helicity_matrix_element.png`, `build_matrix_doc.py` + `doclib.py` (rebuild).